**GENERAL NUMBER PLATE DETECTION PROGRAM**

This number plate detecting program uses only image processing techniques, not AI or deep learning, neural network or Ai based image processing libraries. Main idea is finding rectangle area with many edges and correct size, then confirm it.to detect number plates, I used “Traffic Control CCTV.mp4” video.

In number plate detecting part, I used several techniques,

- Contrast enhancement (CLAHE)
- Noise removal (Bilateral filter)
- Edge detection (Sobel)
- Thresholding
- Morphology
- Contour detection
- Shape filtering
- Edge density & variance checking
- Confidence scoring
- Overlap suppression

In this code, Video read frame by frame, video is resized to fixed height for faster processing and also I show preprocessing videos to get the idea how code works. Then Convert frame to grayscale because color not needed and it reduce processing power needed, simple and fast. Enhance contrast using CLAHE, good for dark or shadowed platescape divide image to small blocks and stretch contrast locally. Reduce noise using Bilateral filter, keep edges sharp for plate borders.

Detect edges using Sobel operator in X and Y directions. Threshold gradient image to get strong edges, weak edges removed. Then Apply morphological closing with rectangular kernel to join broken edges.

Find contours from binary image, create bounding box for each Filter candidates using width, height, aspect ratio and area to match plate shape Approximate contour as polygon, keep rectangles with 4–6 points.

Validate candidate using edge density and variance Plates have many edges and brightness changes, flat regions rejected. Increase confidence score if candidate has more edges or variance, max value is 1.0.Remove overlapping candidates using simple non-maximum suppression. Keep only high confidence plates. Then Draw rectangle on final frame to show detected plates.

General Number Plate Detection program process  flow chart (Part 01)

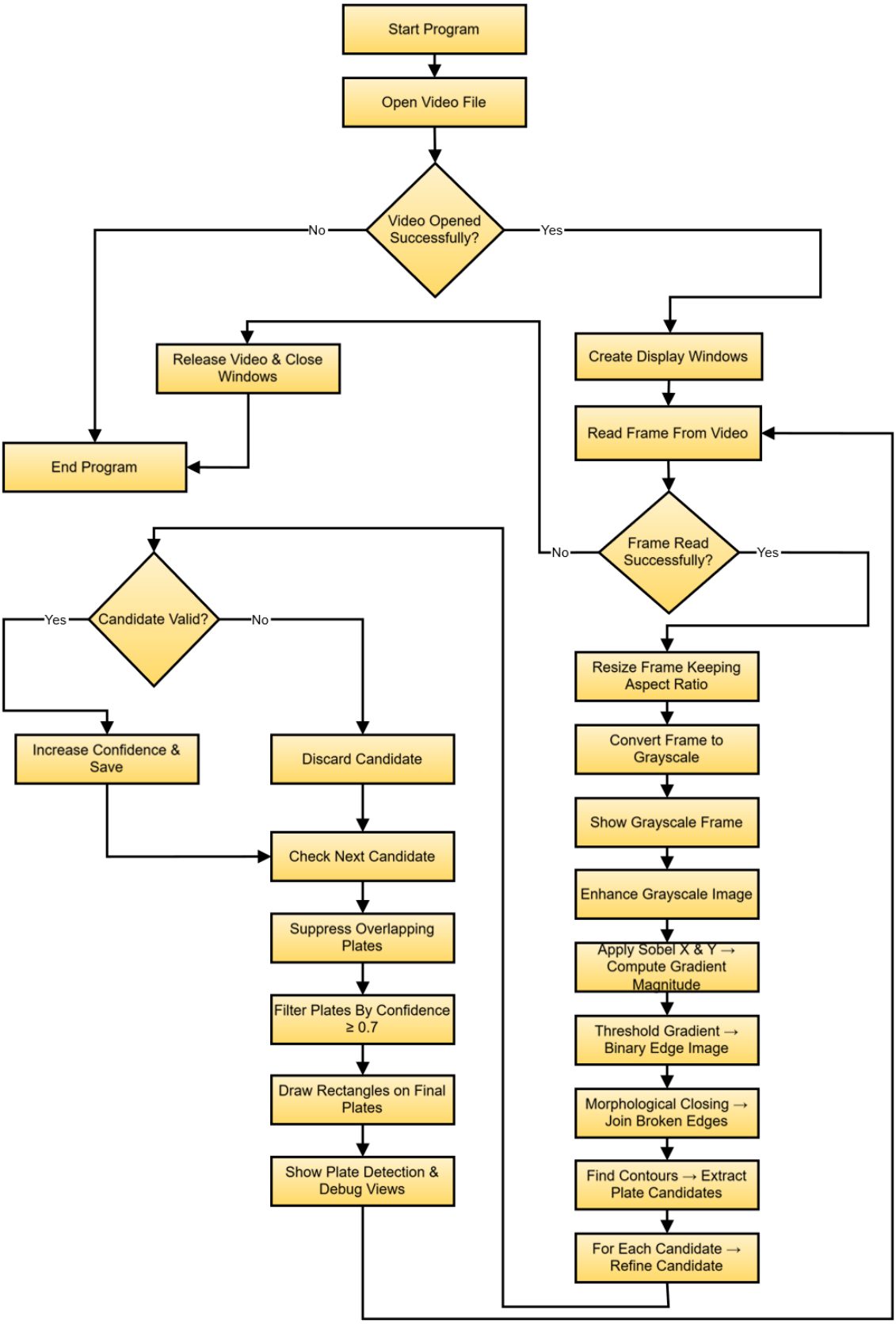


*Figure 1 : General Number Plate Detection program process flow chart*

The main video window shows the original input video with detected number plates highlighted using bounding boxes. The additional preprocessing windows display intermediate processing stages, including grayscale conversion, contrast enhancement using CLAHE, edge detection, thresholding, and morphological operations. These preprocessing views help to visualize how the algorithm enhances plate features, suppresses noise, and extracts candidate regions before final number plate detection.

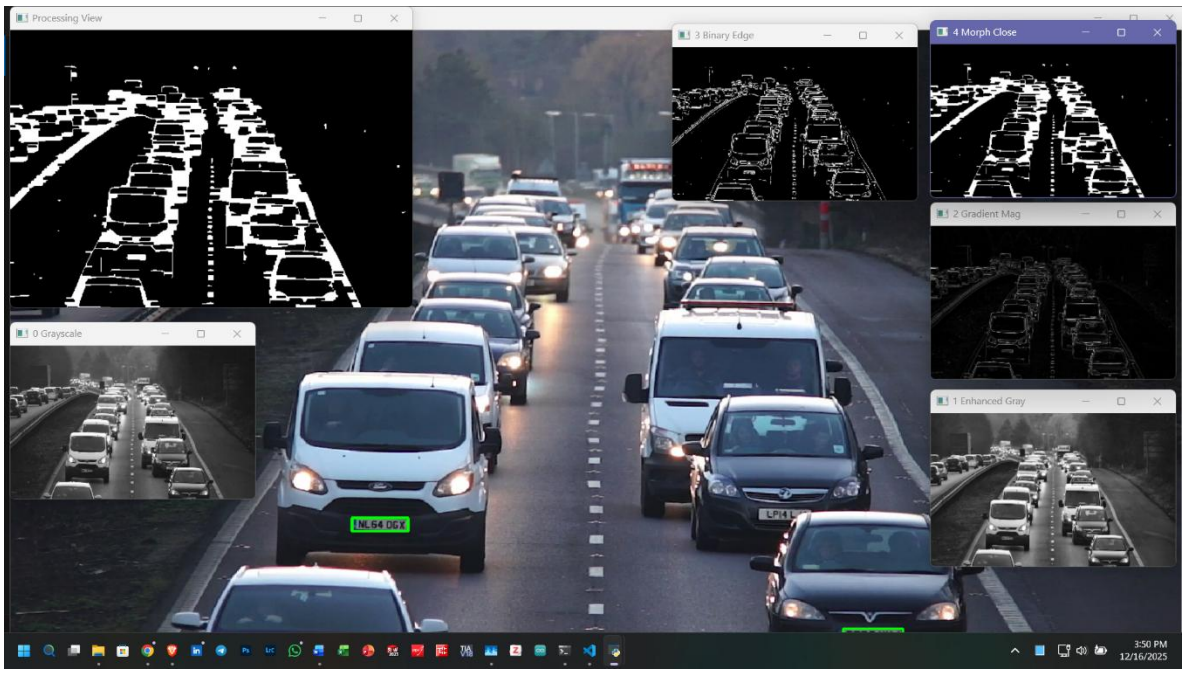


*Figure 3 : general number plate detection process sample 1*

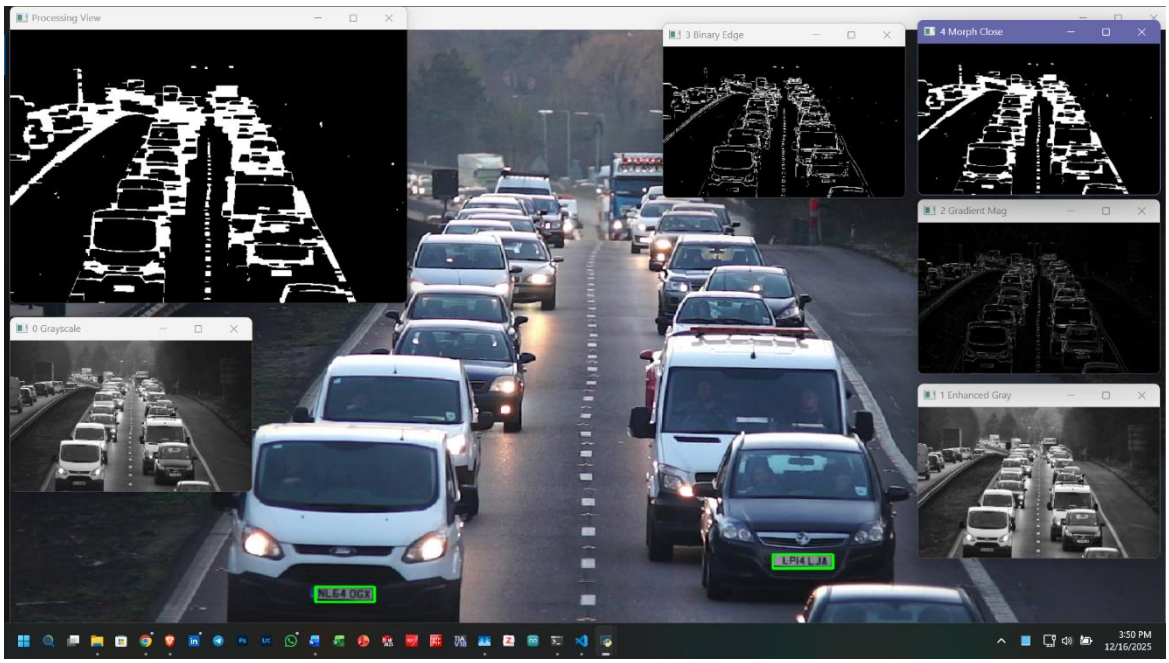


*Figure 2 : general number plate detection process sample 2*

**SPECIFIC NUMBER PLATE DETECTION PROGRAM (PART 2)**

This program develops from Part 1 to detect a specific number plate using a template image. It still uses only classical image processing, no AI, deep learning or neural networks. Main idea is first detecting possible plates using same method as Part 1, then check if any plate matches the given template.

In this code, I used all techniques from Part 1 and added template matching. The techniques are:

- Contrast enhancement (CLAHE)
- Noise removal (Bilateral filter)
- Edge detection (Sobel)
- Thresholding
- Morphology
- Contour detection
- Shape filtering
- Edge density & variance checking
- Confidence scoring
- Overlap suppression
- Template matching for specific plate

In this part, Video is read frame by frame, resized to fixed height for faster processing. Convert frame to grayscale to reduce processing like part 01 and also template matching works on single channel images. Enhance contrast with CLAHE and reduce noise with Bilateral filter to keep edges clear.in this part, I remove code lines that are related to show preprocessing frames to simplify the code.

Detect edges using Sobel in X and Y, calculate gradient magnitude, threshold to get strong edges only, apply morphology to join broken edges. Find contours, filter candidates by size, aspect ratio, area, polygon approximation. Check edge density and variance, increase confidence for better quality, remove overlapping plates.

Then for each candidate plate, apply multi-scale template matching with given template image. Template is resized 40% to 200% to match different plate sizes. Then Compare template with candidate region using normalized correlation coefficient. If score higher than threshold, plate is specific plate and draw red rectangle for specific plate to show that specific plate, green for others number plates. Finally show detected plates on frame. Program can exit by pressing Q.

Specific number plate detection program process flow chart

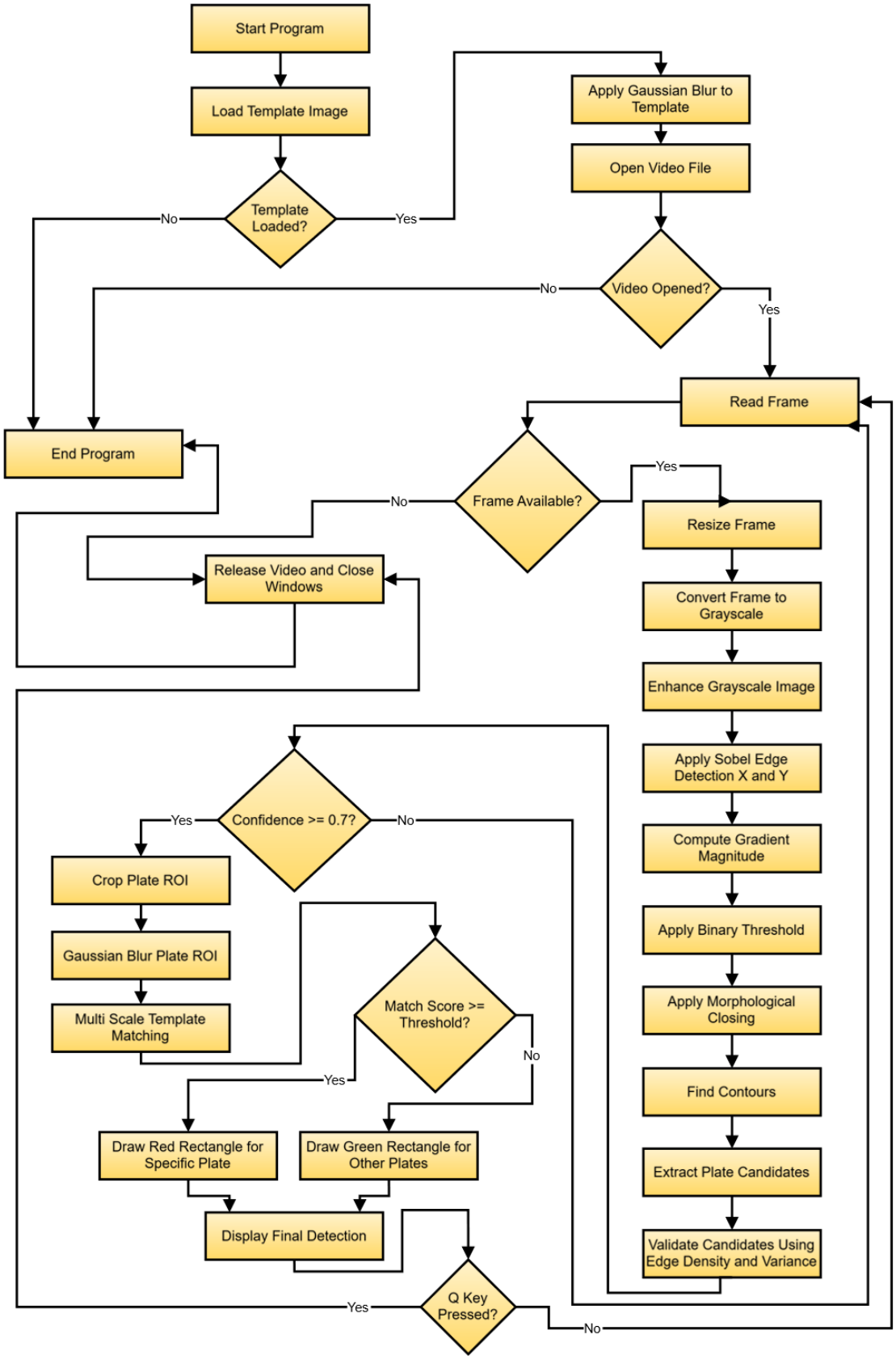


*Figure 4:Specific number plate detection program process flow chart*

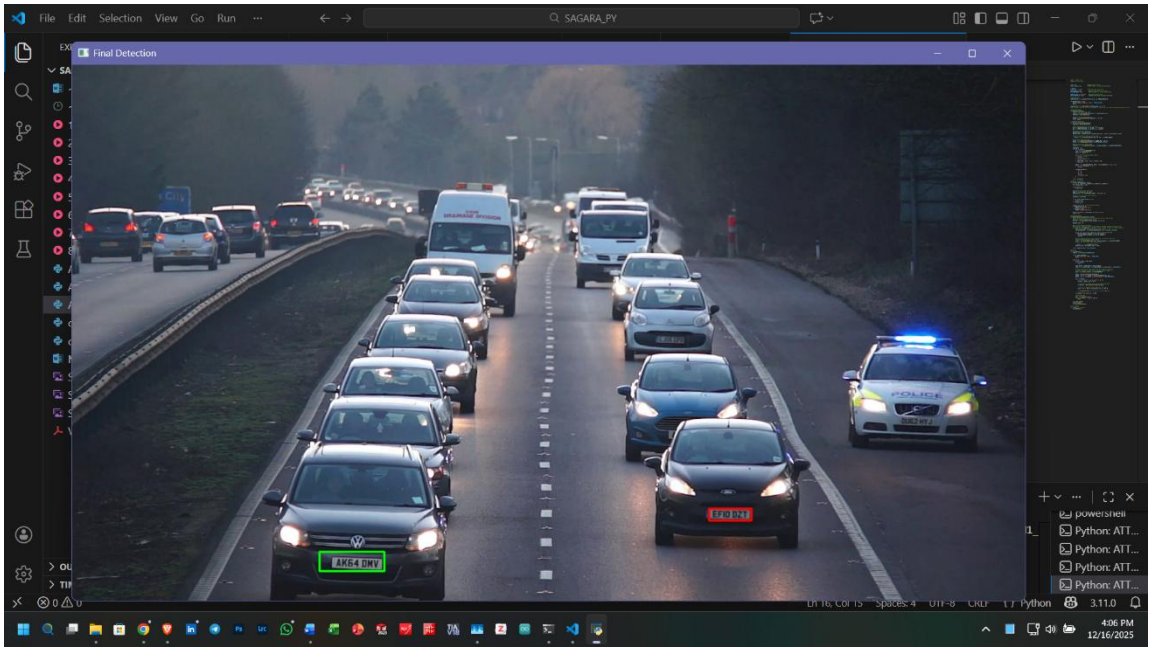


*Figure 5 : Specific number plate detection program process sample 1*

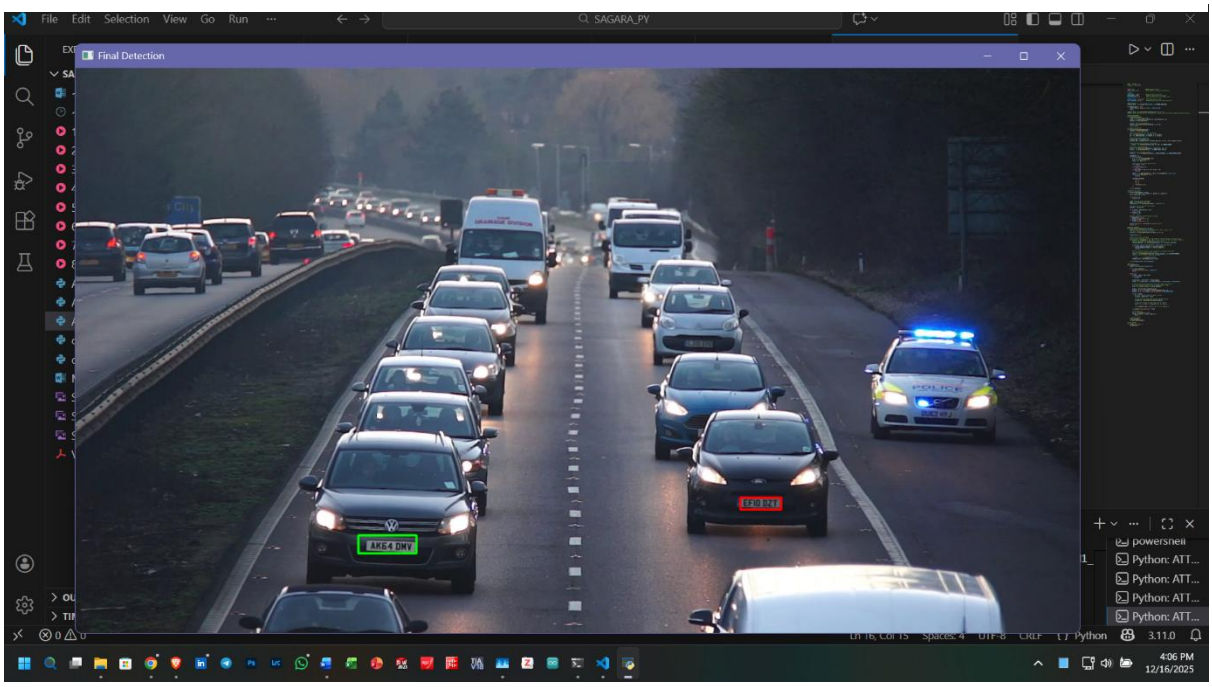


*Figure 6 : Specific number plate detection program process sample 2*

*Figure 7 : Specific Number plate sample photo*